



AI-Assisted Insights for Indian Crop Production

1997–2015 · Evidence-based analysis for smarter agricultural policy

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Can Business Intelligence Help Improve Agricultural Planning in India?

- ❏ **Business Question:** How can historical crop production data help policymakers make better agricultural decisions?



Understanding the Dataset

Indian Crop Production Dataset · 1997–2015 · 238,838 observations

33

States & UTs

19

Years

105

Crops

238K

Records

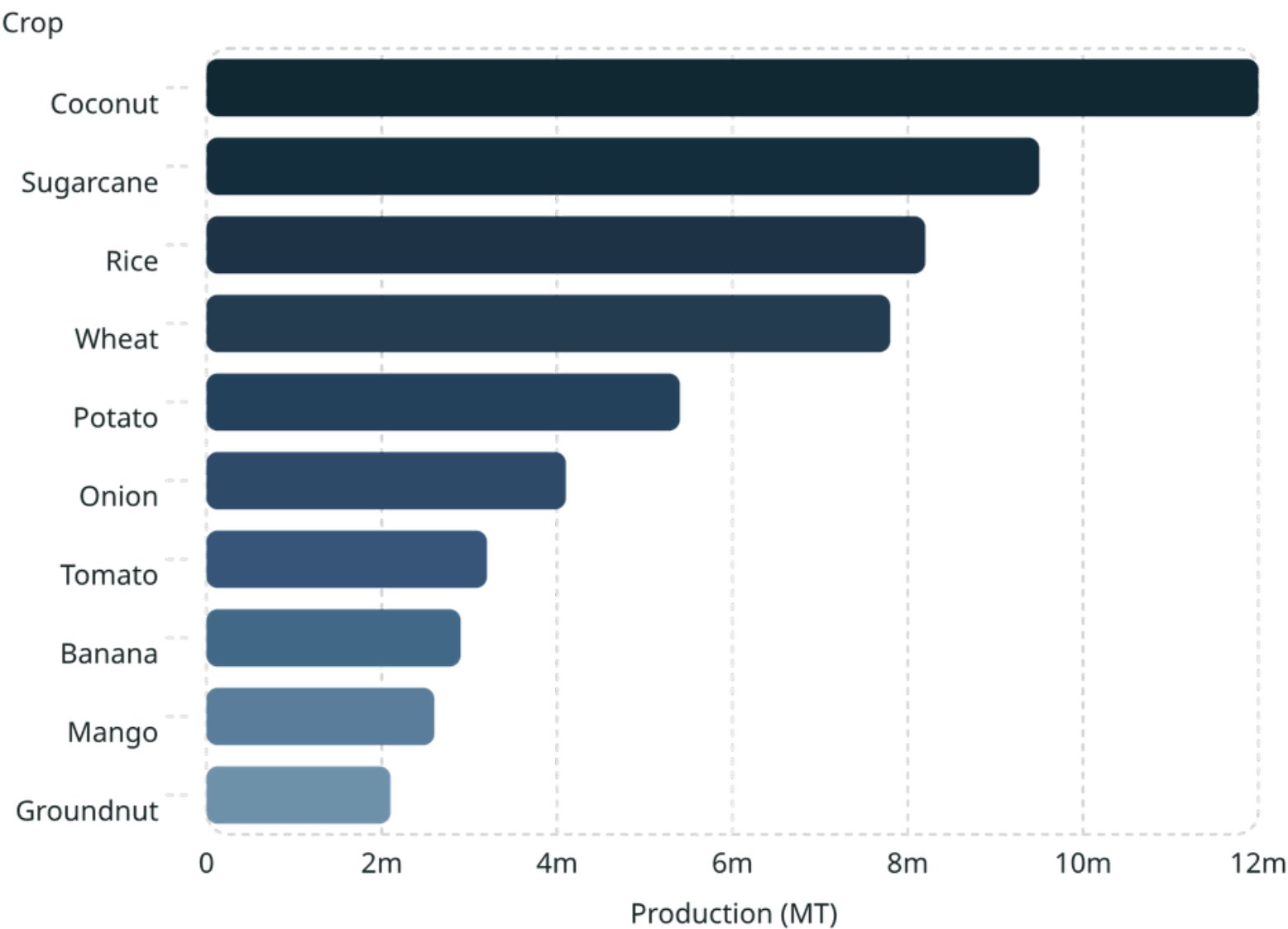
Key Dimensions

- 646 Districts across India
- Multiple growing seasons per year

Core Variables

- Crop · Production · Area · Yield
- State · Season · Year

Crop Production is Concentrated Among a Few Major Crops



What This Tells Us

Coconut dominates total production volume. The steep drop-off across the remaining crops signals strong concentration — a critical risk factor for agricultural resilience.

High production concentration increases dependency on a limited number of crops.

Agricultural Production Changes Over Time

Long-term trend analysis supports better forecasting and resource planning.

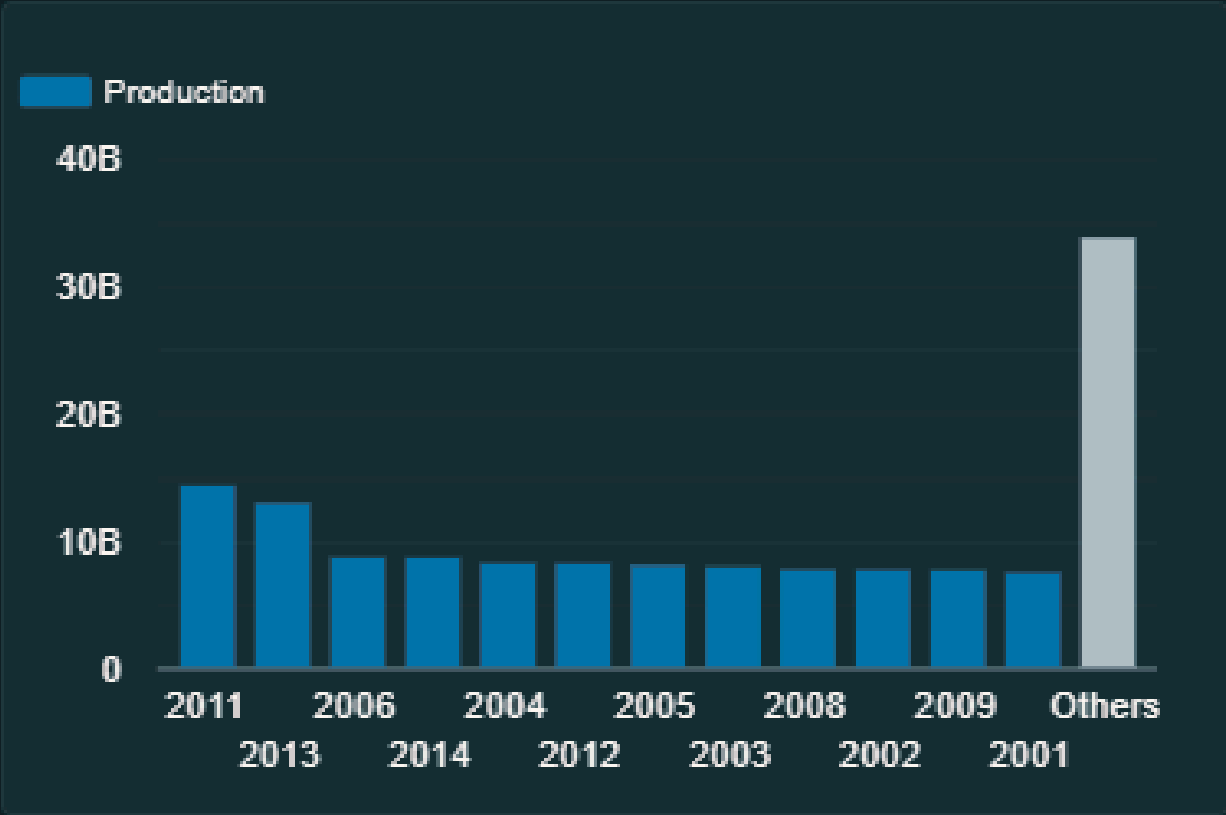
Total Production
141,176,116,767.39

Total Area
2,947,831,968.25

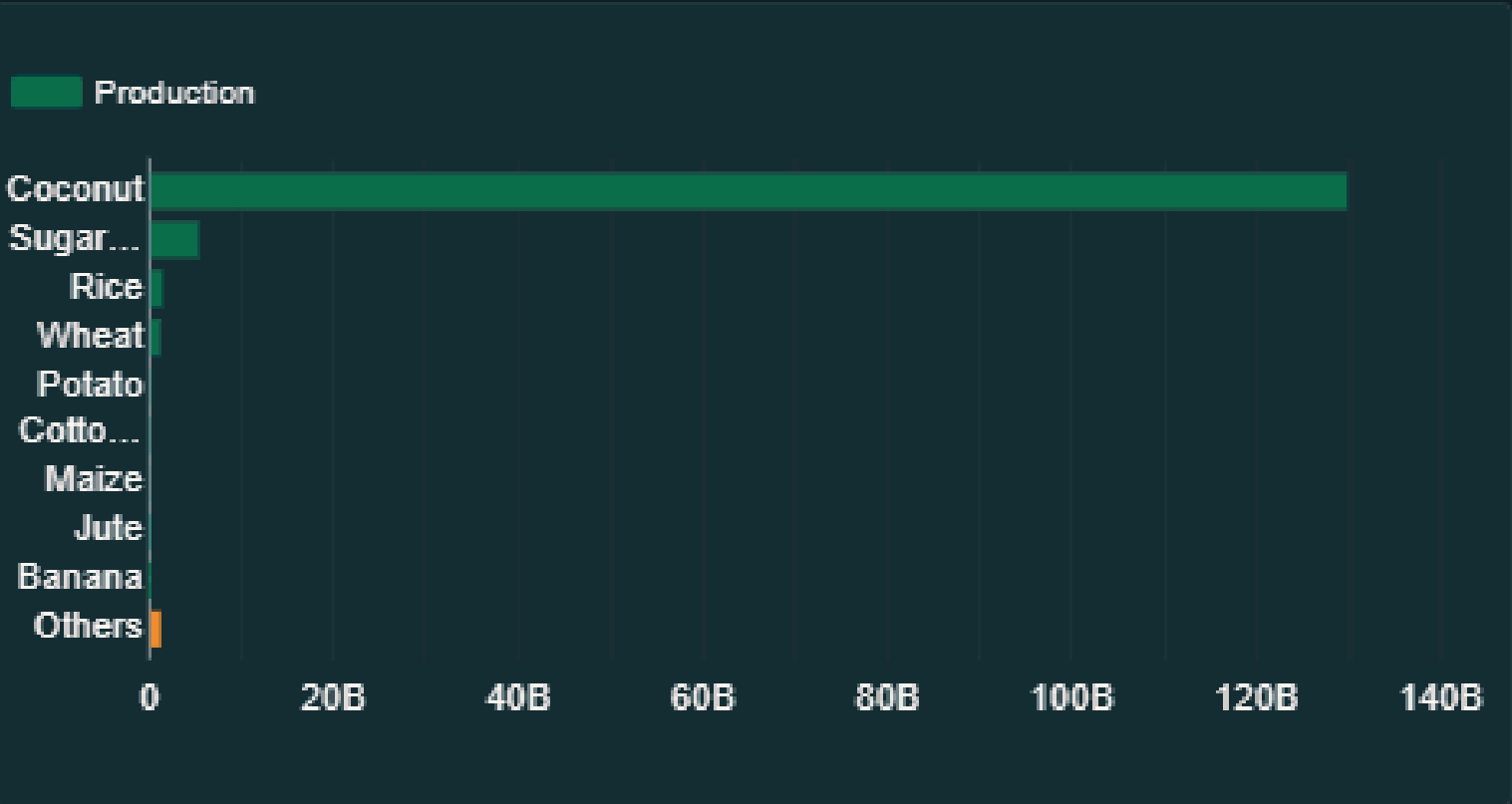
Number of Cr...
105

Average Yield
42.26

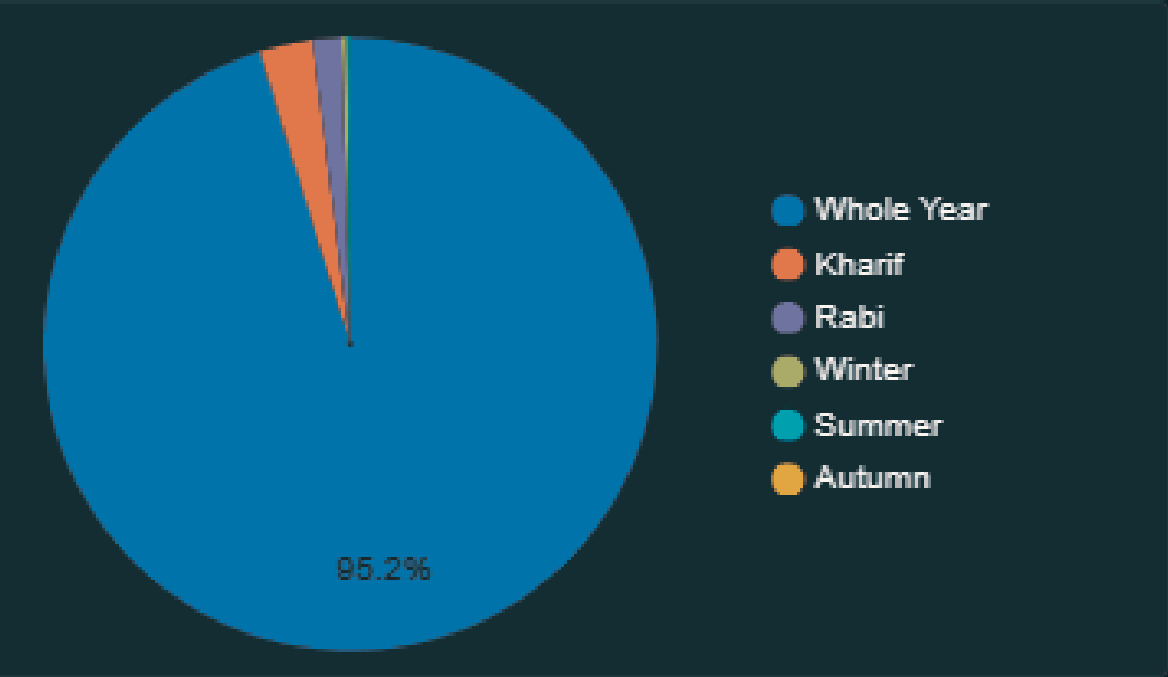
Production in timeline (2000-2015)



Top 10 Crops by Production



Production Share by Season

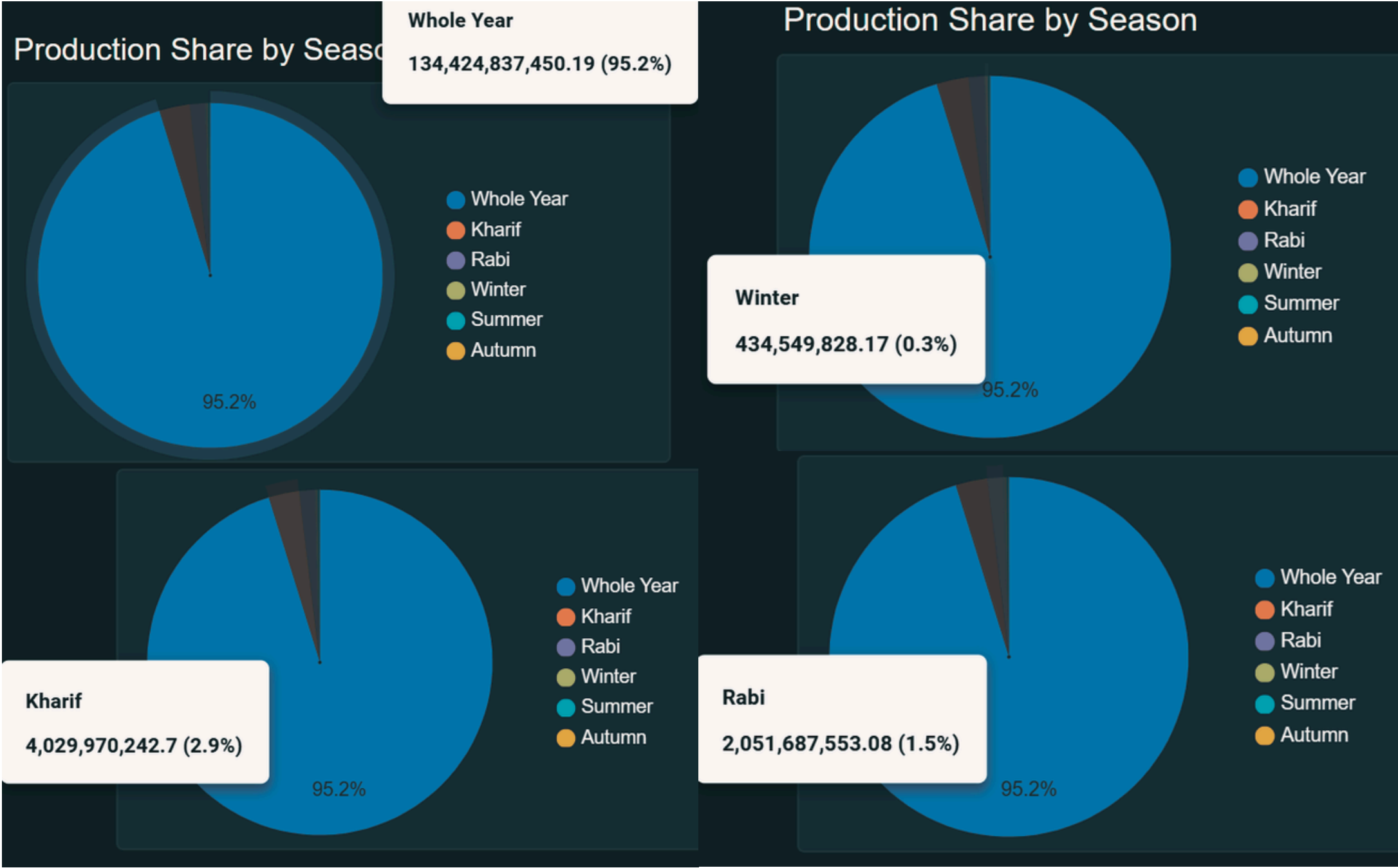


Select State

Select Year

Seasonal Patterns Influence Agricultural Output

Seasonal insights enable more effective agricultural planning.





Strategic Recommendations

1

Invest in High Performers

Increase investment in lower-producing crops through research, irrigation improvements, and farmer training to reduce dependency on a few major crops.

2

Support Underperforming Crops

Continue supporting high-performing crops while encouraging sustainable farming practices.

3

Deploy Production Analytics

Adopt Business Intelligence dashboards integrated with AI to continuously monitor agricultural production and support evidence-based policymaking.